

Challenge (Habanero)

- Q1.** If $f(x) = x + 2$ and $g(x) = 2x - 1$, find $f(g(2))$.
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9
- Q2.** Determine the domain of $f(g(x))$ where $f(x) = \frac{1}{x}$ and $g(x) = x - 3$.
(A) $(-\infty, 3) \cup (3, \infty)$
(B) $(-\infty, 3)$
(C) $(3, \infty)$
(D) $[3, \infty)$
(E) $(-\infty, 3]$
- Q3.** A construction worker needs to add a 10% surcharge to a \$200 job, then apply a 5% discount. If $f(x)$ represents the surcharge and $g(x)$ represents the discount, find $g(f(200))$.
(A) \$207
(B) \$208
(C) \$209
(D) \$210
(E) \$211
- Q4.** Given $f(x) = 2x$ and $g(x) = x^2 + 1$, evaluate $f(g(2))$.
(A) 10
(B) 12
(C) 14
(D) 16
(E) 18
- Q5.** Find the value of x such that $f(g(x)) = 0$, given $f(x) = x - 1$ and $g(x) = x + 2$.
(A) -3
(B) -2
(C) -1
(D) 0
(E) 1
- Q6.** Determine the range of $f(g(x))$ where $f(x) = x^2$ and $g(x) = 2x - 1$.
(A) $[0, \infty)$
(B) $(-\infty, 0]$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $(-1, 1)$
(E) $[1, \infty)$
- Q7.** A chef needs to cook 3 batches of 24 cupcakes. If $f(x)$ represents the number of batches and $g(x)$ represents the number of cupcakes per batch, find $g(f(3))$.
(A) 24
(B) 48
(C) 72
(D) 96
(E) 120
- Q8.** Find the value of $f(g(0))$ where $f(x) = 2x + 1$ and $g(x) = x - 4$.
(A) -7
(B) -6
(C) -5
(D) -4
(E) -3

Open Ended Questions (FRQ)

- Q9.** A construction company is building a skyscraper with 50 floors. Each floor requires 2000 feet of steel beams. If $f(x)$ represents the number of floors and $g(x)$ represents the amount of steel needed per floor, evaluate $g(f(50))$ and provide the units for the answer.
- Q10.** A recipe for making cakes requires 2 cups of flour per cake. If $f(x)$ represents the number of cups of flour and $g(x)$ represents the number of cakes, determine $f(g(10))$ and explain the real-world meaning of the result.

Chef's Tips

- Tip 1: Encourage students to use diagrams to visualize composite functions.
- Tip 2: Remind students to consider domain restrictions when evaluating composite functions.
- Tip 3: Emphasize the importance of interpreting compositions in real-world contexts.